Labsheet-09

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```
!pip -q install ultralytics
```

```
                                        46.6/46.6 kB 1.2 MB/s eta 0:00:00
                                        1.4/1.4 MB 22.7 MB/s eta 0:00:00
                                        53.2/53.2 kB 3.4 MB/s eta 0:00:00
                                        77.7/77.7 kB 4.6 MB/s eta 0:00:00
```

```
import cv2
import matplotlib.pyplot as plt
from ultralytics import YOLO

print("Libraries imported successfully!")
```

```
Libraries imported successfully!
```

```
model = YOLO("yolov8n.pt")
print("YOLOv8 model loaded successfully!")
```

```
YOLOv8 model loaded successfully!
```

```
!wget -q https://ultralytics.com/images/bus.jpg -O bus.jpg
print("Image downloaded successfully!")
```

```
Image downloaded successfully!
```

```
image = cv2.imread("bus.jpg")
image = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)

plt.imshow(image)
plt.axis("off")
plt.show()
```

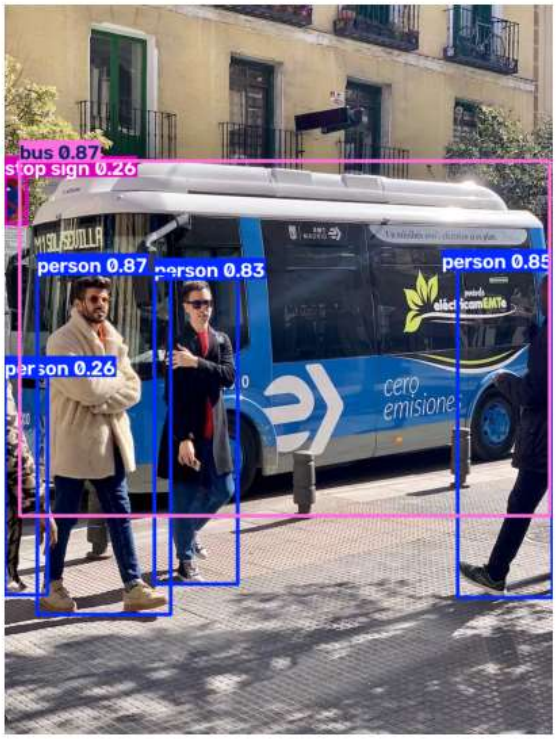



```
results = model("bus.jpg")
print("Object detection completed!")
```

```
image 1/1 /content/bus.jpg: 640x480 4 persons, 1 bus, 1 stop sign, 496.1ms
Speed: 28.2ms preprocess, 496.1ms inference, 40.2ms postprocess per image at shape (1, 3, 640, 480)
Object detection completed!
```

```
result = results[0].plot()

plt.figure(figsize=(10,7))
plt.imshow(cv2.cvtColor(result, cv2.COLOR_BGR2RGB))
plt.axis("off")
plt.show()
```




```
for box in results[0].boxes:
    cls = int(box.cls[0])
    conf = float(box.conf[0])
    print(model.names[cls], ":", round(conf, 2))
```

```
bus : 0.87
person : 0.87
person : 0.85
person : 0.83
person : 0.26
stop sign : 0.26
```

```
print("Total objects detected:", len(results[0].boxes))
```

```
Total objects detected: 6
```

```
objects = []

for box in results[0].boxes:
    cls = int(box.cls[0])
    objects.append(model.names[cls])

print("Detected objects:")
print(objects)
```

```
Detected objects:
['bus', 'person', 'person', 'person', 'person', 'stop sign']
```